

LangChain Transformers Python

Text summarization and Q&A using **HuggingFace transformer models** integrated with **LangChain** · runs fully local with GPU acceleration.

· Features

- **Text Summarization** · BART-large-cnn for abstractive summaries
 - **Question Answering** · RoBERTa-base-squad2 for QA on summaries
 - **Configurable Length** · Short, medium, or long summaries
 - **GPU Acceleration** · CUDA device 0 support
 - **LangChain Integration** · PromptTemplate and chain syntax
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.. Architecture

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User Text · BART Summarizer · Summary
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User Question · RoBERTa QA Model .. · Answer
```

Models Used

Model	Task	Source	----- ----- -----	facebook/bart-large-cnn	Summarization	HuggingFace	
facebook/bart-large	Refinement	HuggingFace		deepset/roberta-base-squad2	Question Answering	HuggingFace	

· Getting Started

```
cd Langchain-Transformers-Python
pip install -r requirements.txt
python main.py
```

· Logic Flow

- 1 **Input** · User enters text to summarize
 - 2 **Summarize** · BART-large-cnn generates abstractive summary
 - 3 **Refine** · Optional refinement pass with BART-large
 - 4 **Display** · Summary shown to user
 - 5 **Q&A Loop** · User asks questions about the summary
 - 6 **Answer** · RoBERTa extracts answers from the summary context
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· Dependencies

transformers, langchain, huggingface-hub, torch

· License

Educational project · use freely for learning and reference.